# E-commerce SQL Pipeline

A reproducible MySQL analytics pipeline — data modeling, cleaning, analysis, and advanced SQL features.

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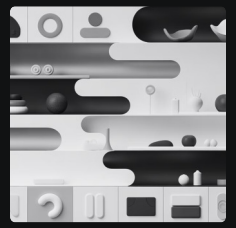
# The Dataset

### country_raw.csv

Country codes & names (with duplicates)

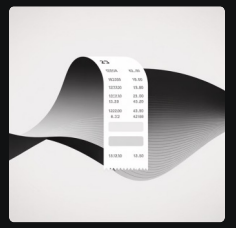
### customer_raw.csv

Customer ID, name, gender, birthdate, country



### product_raw.csv

Product ID, name, category, price



### sales_raw.csv

Customer, product, date, quantity, amount

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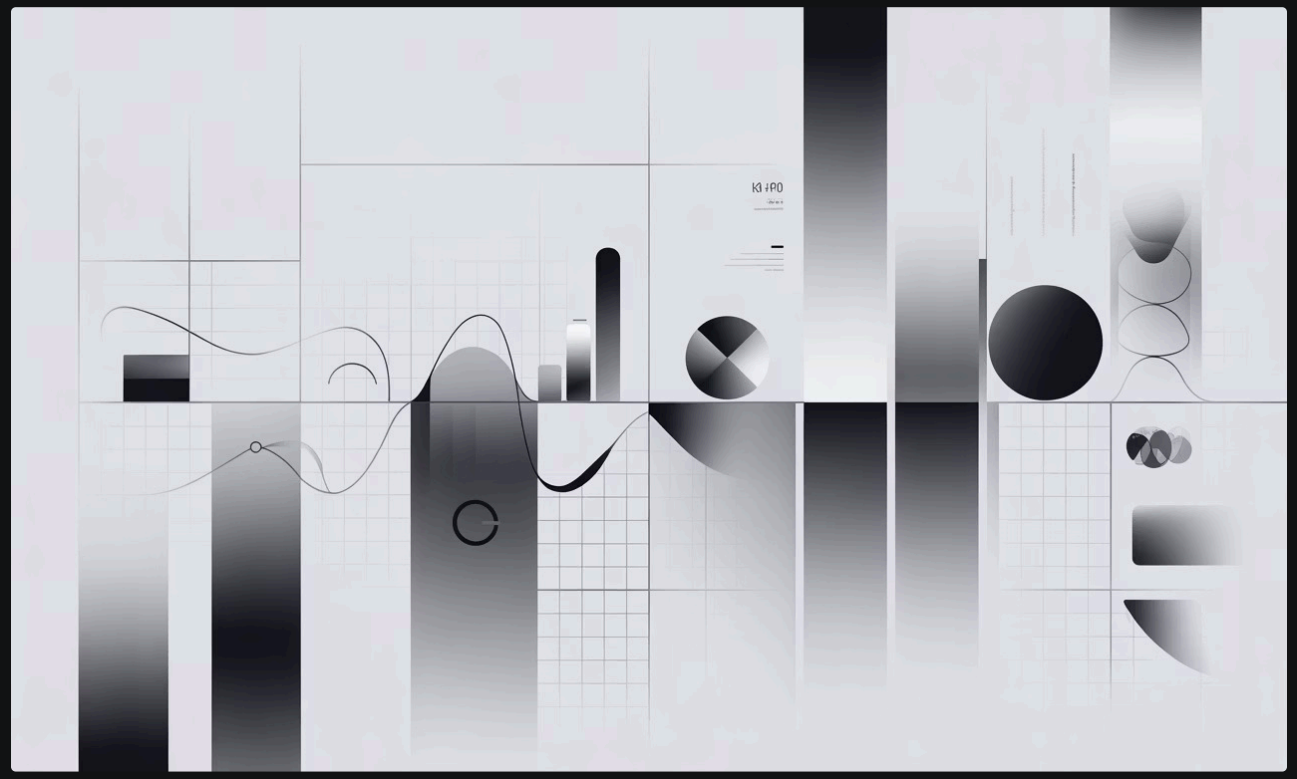
# Business Questions



### Revenue Trends

Monthly revenue evolution

### Top Performers

Best products & customers by sales

### Category Insights

Average ticket by product category

### Geographic Breakdown

Sales by country & month

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# Project Structure

| | |
|---|---|
| 1 | **01_schema.sql**<br>Creates staging & core tables |
| 2 | **02_load_staging.sql**<br>Data quality checks after CSV import |
| 3 | **03_transform_core.sql**<br>Cleans & loads core tables |
| 4 | **04_semantic_views.sql**<br>Business views for analysis |
| 5 | **05_analysis_queries.sql**<br>8+ documented analytical queries |
| 6 | **06 & 07**<br>Quality checks & advanced SQL |

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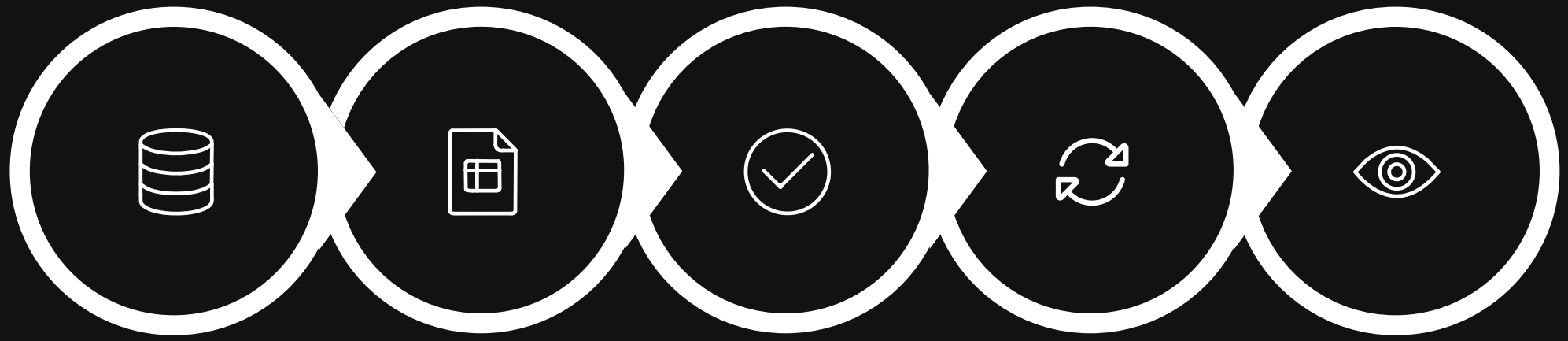
# How to Reproduce



**Create DB** → **Import CSVs** → **Validate Import** → **Transform & Load** → **Business Views**

Compatible with MySQL Workbench, DBeaver, or CLI. Tested on MySQL 8+.

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# Data Quality Findings

## Duplicate Countries

Duplicate codes found in country_raw.csv — caught at staging

## Missing Fields

Some customers lack country codes or have malformed birthdates

## Invalid Numerics

Products & sales with invalid values filtered during transformation

## Quality Checks

No nulls in keys, no orphaned FKs, no negative sales

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# Advanced SQL Features

### Safe % Function

Division-safe percentage calculation

### Refresh Procedure

Core reload with transaction & error handling

### Audit Trigger

Logs every sales insertion automatically

### Smoke Test

Validates full pipeline health end-to-end

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# Assumptions & Limitations

### MySQL Only

Syntax tested on MySQL 8+ — no cross-DB compatibility

### CSV Import As-Is

All cleaning done in SQL; raw files are never edited

### Known Issues Flagged

Missing country codes are flagged, not fixed

### Clarity Over Scale

Designed for reproducibility, not large-scale performance

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